Overview of Kalman Filter Theory and Applications

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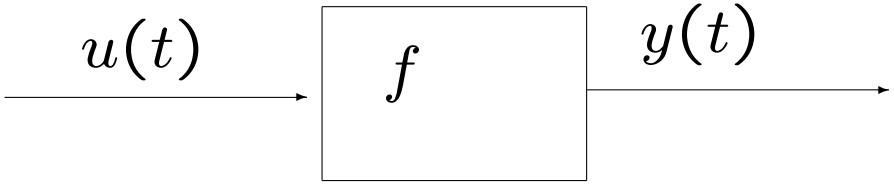
Segment 3
Differential Equations and Difference Equations
Topics

- Input-Output Representation

- Differential Equations and Difference Equations

Input-Output Representation

- Black box approach: $y(t) = f(u(t), t)$



- Output represents the resultant behavior of the dynamic system as a function of the input (and possibly time).

- Output is often one primary scalar variable of interest like position.

Input-Output Representation

- Note that output is not necessarily the same thing as the observation or measurement.

- Observation or measurement is often related directly to the output through a matrix coupling.

- In state space representation, the output is often eliminated in favor of the state vector, and the measurement is given as a function of the state vector directly.

Input-Output Representation

- The derivatives of position and how they relate to the forcing function yields a single differential equation.

- We'll see later how an nth-order, scalar differential equation is related to a vector state space model.

Differential Equations and Difference Equations

- Ordinary Differential Operators
- Ordinary Differential Equations
- Difference Equations

Ordinary Differential Operators: Vectors and Functions

- Vectors are functions and functions are vectors
- A vector $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ in 3-space is really a function $x : \{1, 2, 3\} \to \mathbb{R}$
- We just write them differently than functions, i.e., instead of $x(i)$ we write x_i.

Ordinary Differential Operators: Vectors and Functions

- And since the domain $\{1, 2, 3\}$ of the function x is so small, we can exhibit all values of x explicitly in one fell swoop, i.e., as $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

- But, by the same token, ordinary real functions $f : \mathbb{R} \to \mathbb{R}$ are vectors

- They just have so many components that we can't exhibit them all at once

Ordinary Differential Operators: Vectors and Functions

- We can add vectors by adding their components: $z = x + y$ if and only if $z_i = x_i + y_i$ for each index i

- By the same token, we can add functions componentwise: $h = f + g$ if and only if $h(t) = f(t) + g(t)$ for all t in the domain

- Componentwise multiplication of functions can also be defined (although this does not come up as often for ordinary vectors): $h = f \cdot g$ if and only if $h(t) = f(t) \cdot g(t)$ for all t in the domain

Ordinary Differential Operators: Vectors and Functions

- Finally, we can multiply functions by scalars, just like we do for vectors: $g = a \cdot f$ if and only if $g(t) = a \cdot f(t)$ for all $t \in \mathbb{R}$.

- The upshot of all this is that the set of all real functions is a vector space!

- But we are really more interested in a subspace of this vector space, i.e., we really want to restrict our attention to the set of all **differentiable** functions on some closed interval.

Ordinary Differential Operators: Differentiable Functions

- Let $I = [a, b]$ be any closed interval of the real line
- Consider the set $\mathbb{R}^I = \{f : I \to \mathbb{R}\}$ of all functions mapping $I = [a, b]$ into the real numbers
- Then, as we have just pointed out, $\mathbb{R}^I$ is a vector space over the reals

Ordinary Differential Operators

- We define the **derivative** of a function f at the point $t \in [a, b]$ as the limit as $s \longrightarrow t$ of the difference quotient,

$$\frac{f(s) - f(t)}{s - t},$$

  assuming this limit exists.

- Note that not all functions have derivatives!

- Those that do have a derivative defined at every point of I we call **differentiable** on I.

Ordinary Differential Operators

- This operation turns out to be a linear operator on the vector space of functions that are differentiable on $I = [a, b]$.

- In fact, we want to consider those functions that can be differentiated as many times as we like

- Thus, we consider the space of infinitely differentiable, real-valued functions defined on the closed real interval $[a, b]$: $C^\infty[a, b]$

Ordinary Differential Operators

- The operation of taking the derivative will enable us to develop deterministic differential equations.

- Later we will define derivatives of stochastic processes in order to introduce and solve stochastic differential equations. You cannot always take the derivative (or even integrate) a random process – and sometimes not even the sample functions of a random process – in the ordinary sense.

Ordinary Differential Operators

- Since the derivatives of all orders are assumed to exist for all functions $f \in C^\infty[a,b]$, the limit of the difference quotient always exists for such functions.

- Thus, we assign to each f a new function Df given by

$$[Df](t) = \lim_{s \to t} \frac{f(s) - f(t)}{s - t} \tag{1}$$

- This assignment defines a transformation

$$D : C^\infty[a,b] \to C^\infty[a,b]$$

that is a linear operator.

Ordinary Differential Operators

- D is linear: $D[af + bg] = a[Df] + b[Dg]$

- Sometimes we get lax and simply write $Df(t)$ instead of $[Df](t)$. However, keep in mind that $Df(t)$ says "Differentiate f and then evaluate the resulting function at the point t."

- D obeys all the usual rules of differential calculus such as the product rule and the chain rule:

$$D(fg) = [Df]g + f[Dg]$$

$$D(f \circ g) = [Df] \circ g \cdot Dg$$

Ordinary Differential Operators: Examples

- If id is the identity function, i.e., $\mathrm{id}(t) = t$, then $D\mathrm{id} = 1$. (You probably know this as $\frac{dt}{dt} = 1$; Note that 1 is the function whose value at every t is 1.)

- Using the product rule, we have, for example,

$$\begin{aligned} D\mathrm{id}^2 &= D[\mathrm{id} \cdot \mathrm{id}] \\ &= D\mathrm{id} \cdot \mathrm{id} + \mathrm{id} \cdot D\mathrm{id} \\ &= 1 \cdot \mathrm{id} + \mathrm{id} \cdot 1 \\ &= \mathrm{id} + \mathrm{id} \\ &= 2\mathrm{id} \end{aligned}$$

  (In more familiar notation this is just $\frac{dt^2}{dt} = 2t$.)

Ordinary Differential Operators

- Note that we sometimes write d/dt or $\frac{d}{dt}$ for the operator D when we want to emphasize the independent variable t. Or, instead of $\frac{dy}{dt}$ we write simply $\dot{y}$.

- However, that is really unnecessary, adds no new information, and is just extra baggage.

- Nevertheless, we shall do it sometimes because it may be more familiar.

Ordinary Differential Operators

- Since there is really only one generating operator D that we are interested in here, we are less interested in multiplication than we are in function composition or iteration of this operator. For instance, $D^2 f = [D \circ D](f) = D(Df)$.

- By induction we have $D^n = \underbrace{D \circ \cdots \circ D}_{n}$.

- Just as real-valued functions inherit algebraic operations from those of their range spaces, so too do differential operators:

$$[aD^m + bD^n](f) = a(D^m f) + b(D^n f)$$

Ordinary Differential Operators

- We can therefore generate polynomials in D:

$$p(D) = a_n D^n + \cdots + a_1 D + a_o I,$$

where I is the identity operator on $C^\infty[a, b]$. We usually just drop the identity operator, since it amounts to multiplying by 1.

- Such polynomials in D are still linear operators.

- Thus, we have rediscovered the group ring $\mathbb{R}$(Diff), where Diff in this case is the infinite cyclic group generated by the differential operator D. We'll call this group ring the **polynomial ring of differential operators**.

Ordinary Differential Equations

- With polynomials of differential operators we can now define ordinary differential equations.

- An equation of the form

$$p(D)f = q(D)g$$

where $f, g \in C^\infty[a, b]$ and $p(D), q(D) \in \mathbb{R}(\text{Diff})$, is called an ordinary differential equation with constant coefficients.

Ordinary Differential Equations

- It is probably more familiar if we use y instead of f and u instead of g, and if we show the independent variable, say t, explicitly:

$$a_n\frac{d^n y}{dt^n} + \cdots + a_1\frac{dy}{dt} + a_0 y = b_m\frac{d^m u}{dt^m} + \cdots + b_1\frac{du}{dt} + b_0 u$$

or

$$\sum_{i=0}^{n} a_i\frac{d^i y}{dt^i} = \sum_{i=0}^{m} b_i\frac{d^i u}{dt^i} \tag{2}$$

- For our purposes, the function $u = u(t)$ is called the **input**, while the function $y = y(t)$ is called the **output**. The independent variable t is thought of as time.

Ordinary Differential Equations

- In (2), u is ordinarily assumed to be known or given, while y is unknown and sought.

- Any function y satisfying (2) is called a solution of the differential equation.

- A differential equation such as (2) is called **homogeneous** when the input function u is identically zero for all $t \in [a, b]$.

Ordinary Differential Equations

- Thus far we have only dealt with linear differential equations with constant coefficients.

- We could allow the coefficients to be time-dependent functions. However, for most of our applications, this will not be necessary.

Ordinary Differential Equations: Examples

- From physics we have Hooke's law: the restoring force of a spring is proportional to the distance stretched: $-ky = m\frac{d^2y}{dt^2}$.

- This is just an application of Newton's $F = ma$, where a is the acceleration: $a = d^2y/dt^2$.

- Rewriting this in the form of (2) we have

$$m\frac{d^2y}{dt^2} + ky = 0$$

- Therefore, this is a homogeneous equation of order 2.

Ordinary Differential Equations

- We have used y rather than x as in most physics books, because we want to be consistent with (2).

- It is common to rewrite such equations so that the leading coefficient is 1. Thus we have:

$$\frac{d^2y}{dt^2} + \frac{k}{m}y = 0.$$

- It is easily verified that $\sin \omega_0 t$, where $\omega_0 = \sqrt{\frac{k}{m}}$, satisfies this differential equation.

Ordinary Differential Equations

- Equation (2)

$$\sum_{i=0}^{n} a_i \frac{d^i y}{dt^i} = \sum_{i=0}^{m} b_i \frac{d^i u}{dt^i}$$

  is called **time-invariant**, because the coefficients are constant and also because the independent variable t does not appear explicitly in the equation.

- Note that (2) is linear in terms of both the input and the output, because the polynomial differential operator on each side of the equation is a linear operator.

- This double linearity means, on the one hand, that the output y due to several inputs acting at the same time is equal to the sum of the outputs due to each input acting alone.

- On the other hand, it also means that if y_1 and y_2 are both solutions to the homogeneous version of (2), then so is $c_1 y_1 + c_2 y_2$ where c_1 and c_2 are any real constants.

- A maximal set of n such solutions to the homogeneous equation that are also linearly independent (as vectors) is called a **fundamental set**.

- You can find a fundamental set by solving the **characteristic equation** associated with the differential equation. This equation is nothing other than the original polynomial equation in D, but with D replaced by an unknown such as z.

- Thus, the characteristic equation for (2) is

$$a_n z^n + \cdots + a_1 z + a_0 = 0$$

- We use z because the roots of this equation are in general complex (thanks, once again, to Dr. Gauss).

- In solving the characteristic equation, two cases emerge:

  1. The roots of the characteristic equation are all distinct.

  2. Some roots are repeated, i.e., each root r_i has multiplicity n_i.

- In case 1, a fundamental set is easily constructed:

$$y_1(t) = \exp(r_1 t), \ldots, y_n(t) = \exp(r_n t)$$

where the r_i are the distinct roots.

- In case 2, each root r_i of multiplicity n_i contributes n_i functions to the fundamental set:

$$y_{i1}(t) = \exp(r_i t), y_{i2}(t) = t\exp(r_i t), \ldots, y_{in_i}(t) = t^{n_i - 1}\exp(r_i t)$$

Ordinary Differential Equations

- To fully solve the differential equation, the initial conditions need to be taken into account.

- These are the values of the function and its first $n-1$ derivatives at the start of the time interval.

- We will specialize our interval $[a, b]$ to the case in which $a = 0$ and $b = T$, some unspecified time in the future.

- The **free response** is the solution to the homogeneous equation in which the initial conditions are specified but the input is zero, i.e., there is no forcing function.:

  (3) $$y(0), Dy(0), \ldots, D^{n-1}y(0)$$

- The **forced response** reverses this situation; it allows the input function to be non-zero, but it requires all the initial conditions in (3) to be zero.

- The **total response** is the sum of the free and forced responses.

Ordinary Differential Equations: Example of Free, Forced and Total Responses

- Consider $D^2y + 3Dy - 1y = u$ with initial conditions: $y(0) = 1, Dy(0) = -1$.

- The roots (found using Matlab's roots function) are $r_1 = -3.3028$ and $r_2 = 0.3028$, and they are distinct.

- Therefore, a fundamental set is $\{\exp(-3.3028t), \exp(0.3028t)\}$.

Ordinary Differential Equations: Example of Free, Forced and Total Responses

- With the fundamental set and the initial conditions we can form a set of n linear equations in n unknowns (here, $n = 2$):

$$c_1 y_1(0) + c_2 y_2(0) = y(0)$$
$$c_1 Dy_1(0) + c_2 Dy_2(0) = Dy(0)$$

or

$$1 \cdot c_1 + 1 \cdot c_2 = 0$$
$$-3.3028 \cdot c_1 + 0.3028 \cdot c_2 = -1$$

- The previous linear system is of the form $Ax = b$. Using Matlab's matrix division operator \, the solution is $x = A\backslash b$ or

$$c_1 = 0.2773$$
$$c_2 = -0.2773$$

- The total response can also be decomposed into transient and steady state responses

- The **transient** response is that part of the total response that decays to 0 as $t \to \infty$.

- The **steady state response** is the difference between the total response and the transient response.

Linear Time-Invariant Differential Equations

- Linear Time-Invariant (LTI) Differential Equations
  - Linear: Polynomial differential operators are linear:

$$p(D)[af + bg] = ap(D)f + bp(D)g$$

  - Time-Invariant means the coefficients are constant.
  - Homogeneous solution of corresponding state equation is a matrix exponential. (More on that later when we discuss the state transition matrix.)

Ordinary Difference Equations

- Auto-Regressive Moving Average (ARMA)

- n^{th} derivative corresponds to n-step advance:

$$\sum_{i=0}^{n} a_{n-i} y(k+n-i) = \sum_{i=0}^{m} b_{m-i} u(k+m-i) \tag{4}$$

Ordinary Difference Equations

- The left side of Eq. (4) is the Auto-Regressive part
- The right side of Eq. (4) is the Moving Average part
- Hence, ARMA

Ordinary Difference Equations: Example

- Let $n = 3$, $m = 2$ and consider the difference equation:

$$\begin{aligned} -3y(k+3) + 1.5y(k+2) & \\ +7y(k+1) - 3.2y(k) & \\ = 23u(k+2) - 5u(k+1) + 0.5u(k) & \end{aligned}$$

Segment 4
State Space Dynamics
Topics

- Differential Equations and Dynamic Systems
- Block Diagrams and State Space
- State Transition Matrix

Dynamic Systems

- The time-dependent behavior of physical systems can often be modeled by differential equations.

- Such systems are often called **dynamic systems** or **dynamical systems**.

Dynamic Systems

- Considering Newton's law $F = ma$ once again, we see that the force F is the resultant force acting on the mass m.

- However, the resultant force is simply the vector sum of all the forces, both those internal to the system as well as the input forces that are external.

- The restoring force, $-ky$ in the simple oscillator problem, is a force internal to the system. So we don't consider it an input *per se*.

Dynamic Systems

- The response or output of the system is motion, more specifically, accelerated motion given by a.

- Thus, the simple harmonic oscillator is a dynamic system, one in which there is only an internal force determining system behavior: $F = -ky$.

- An external forcing function may also be present, e.g.:

$$\frac{d^2y}{dt^2} + \frac{k}{m}y = \sin \omega t$$

Dynamic Systems

- The most general second-order equation for a mechanical system of this sort according to Newton's law is:

(5) $$\frac{d^2y}{dt^2} = \frac{1}{m}F(y, \dot{y}, t)$$

- If we can separate the forcing function $F(y, \dot{y}, t)$ into a sum of two functions $F_1(y, \dot{y})$ and $F_2(t)$, then we can rearrange this equation into the input-output form of equation (2):

$$\frac{d^2y}{dt^2} - \frac{1}{m}F_1(y, \dot{y}) = \frac{1}{m}F_2(t)$$

Dynamic Systems

- In engineering applications, we are often interested not only in modeling the dynamics of the system, but also in measuring the output or behavior of the system.

- But the output is not always directly observable.

- Thus, in specifying dynamic systems, we often add a second equation, usually algebraic or functional in nature, that relates the output y (and possibly the input u) to some measured or observed quantities.

$$z = h(y) + g(u)$$

The Concept of State

- To determine the behavior of a deterministic system at any future time requires the knowledge of two things:

  1. The equations governing the dynamics of the system

  2. The initial conditions

- But the initial conditions for an nth-order continuous dynamic system are simply the values of the first $n-1$ derivatives at the initial time.

The Concept of State

- At any point in time, you can think of the instantaneous values of the first $n-1$ derivatives as the initial conditions for all future dynamic evolution.

- These values capture the "state" of the dynamic system at that time.

The Concept of State

- Another way of seeing what "state" means is by considering the Taylor series expansion of a function.

- Taylor's theorem says that the value of a function $y(t)$ (if it is sufficiently well-behaved) at any point t close to some nominal point t_0 can be determined from the value of all of its derivatives at the nominal point t_0.

$$y(t) = y(t_0) + \dot{y}(t_0)(t - t_0) + \frac{1}{2!}\ddot{y}(t_0)(t - t_0)^2 + \cdots$$

The Concept of State

- In the language dynamic systems, the quantity $t - t_0$ is simply the elapsed time since the initial time t_0.

- Thus, if you have knowledge of the state at time t_0, i.e., if you know the values of all the (infinitely many) derivatives, then you can determine the output at any later time (sufficiently close to t_0).

- This is because all of the dynamic information is contained in the derivatives.

Dynamic Systems: The Concept of State

- In keeping with the linear spirit of things, we will restrict ourselves for now to linear functions h.

- Also, we will now introduce the state space concept and define z as a function of the state.

- Intuitively, the concept of state involves in some sense the instantaneous internal arrangement of a system. In practical situations, the state variables may represent attributes of the system that we wish to know something about.

Dynamic Systems: The Concept of State

- Mechanical systems like a moving particle have three simultaneous attributes: position, velocity, and acceleration.

- Newton tells us that acceleration is often a function of position, velocity and perhaps time.

- However, as we just saw, a force that acts on a system as a function of time only should be considered an external forcing function, i.e., an input to the system.

- Thus we are led to think of position and velocity as the primary state variables of interest.

State Space Representation

- A general n-th order linear differential equation can be expressed as a system of first-order differential equations.

- Ignoring the input u for the moment, consider:

$$p(D)y = D^n y + a_{n-1}D^{n-1}y + \cdots + a_1 Dy + a_0 y,$$

where we have put the polynomial in normal form, i.e., such that its leading coefficient is 1.

State Space Representation

- If we consider y and its first $n-1$ derivatives as the states of this system, then we have n first order equations. Setting $x_1 = y$, we write

$$\begin{aligned} x_1(t) &= y(t) \\ x_2(t) &= Dy(t) \\ &\cdots \\ x_n(t) &= D^{n-1}y(t). \end{aligned}$$

- The functions $x_i(t)$ are called the **state variables**.

State Space Representation

- Taking the derivative of these n state variables and making use of the original differential equation yields:

$$\begin{aligned} Dx_1(t) &= x_2(t) \\ Dx_2(t) &= x_3(t) \\ &\cdots \\ Dx_n(t) &= -a_0 x_1(t) - a_1 x_2(t) - \cdots - a_{n-1} x_n(t). \end{aligned}$$

- These n equations can be written more compactly in vector matrix form: $Dx(t) = Ax(t)$ or $\dot{x}(t) = Ax(t)$.

State Space Representation

- In the equation $\dot{x}(t) = Ax(t)$, we have:

$$x(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix}$$

and

$$A = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & 0 & 0 & 1 \\ -a_0 & -a_1 & \cdots & -a_{n-2} & -a_{n-1} \end{bmatrix},$$

the **companion matrix** of the differential equation.

State Space Representation

- If there is an input function u and it is present via its derivatives as in equation (2), then we can write:

  (6) $$\dot{x} = Ax + Bu$$

  forming B in much the same way we formed A, and suppressing the time variable t.

- Equation (6) expresses the system's **state dynamics**.

- Later we will include white noise as an input. However, that will make the differential equations stochastic, and we are not yet ready for that.

State Space Representation

- To obtain the original output y as a function of the internal state of the system, we require one additional vector-matrix equation:

  (7) $$y = Cx + Du$$

  where the matrix D represents a **direct feedthrough** of the input to the output.

- The output equation is derived directly from the b_i coefficients in the original matrix differential equation.

State Space Representation

- Kalman filter theory generally assumes $D = 0$ in the output equation and uses a measurement model instead:

  (8) $$z = Hx$$

- Note that H implicitly includes C as a factor.

- Note that we have not placed any restrictions on the size of the measurement column vector z. As long as H has n columns it can have as many rows as desired.

State Space Representation: Example

- Let's consider once again the simple harmonic oscillator:

$$m\frac{d^2y}{dt^2} + ky = 0$$

and convert this to state space form:

$$x_1 = y$$
$$x_2 = Dy$$

and $\dot{x}_2 = -\frac{k}{m}x_1 + 0 \cdot x_2$.

- Therefore, in matrix and vector form:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -k/m & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

- For a measurement model, we could consider a camera with a strobe flash that periodically records the instantaneous position of the sliding mass.

- In this case, only the position would be observable and we would write the measurement or observation model as:

$$z = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

(Remember that we are ignoring both process and measurement noise.)

State Space Representation: Matlab / Simulink Example

- Four parameters A, B, C, D from Matlab workspace
- Measurement model added external to Simulink's state space block

State Space Representation

- State space representation not derived only from single n^{th} order differential equation

- Sometimes state coupling comes more directly through internal dynamic coupling. E.g.,
  - coupled harmonic oscillators
  - complicated RLC networks

Ordinary Difference Equations

- For difference equations, a state space model can be derived using the companion matrix exactly as for differential equations

- Matlab / Simulink has a subsystem block in its Discrete block library in which the four matrices for an input-output model can be specified in the workspace.

State Space Representation: Summary

- The state variables are often provided by the 0th through $n-1^{\text{st}}$ derivatives of a single n^{th} plus internal dynamic coupling

- Equivalently, they are the outputs of the n integrators of the system.

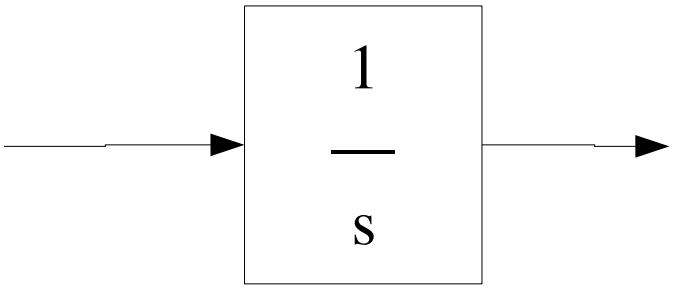
- They represent implicit or explicit internal coupling of the system.

- Similarity transformation of state vector yields equivalent representation. Changing coordinate systems.

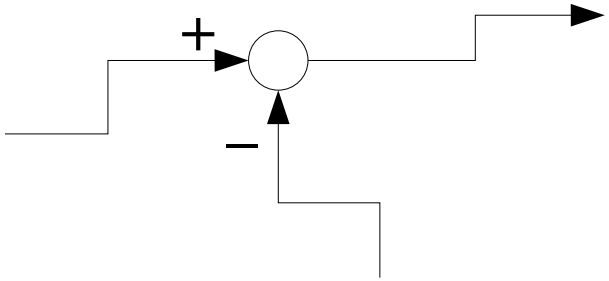
Block Diagrams
depict ...

- System signal flow
  - summation and takeoff points
- Various transformation blocks
  - integrators (continuous) and delayors (discrete)
  - differentiators
  - scalors or gain multipliers

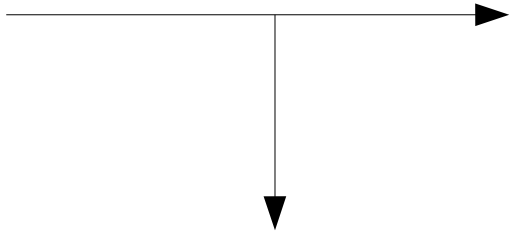
Integration Block



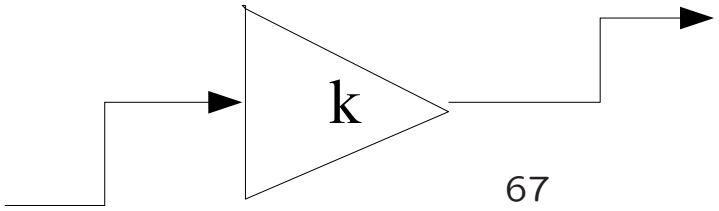
Summation Point

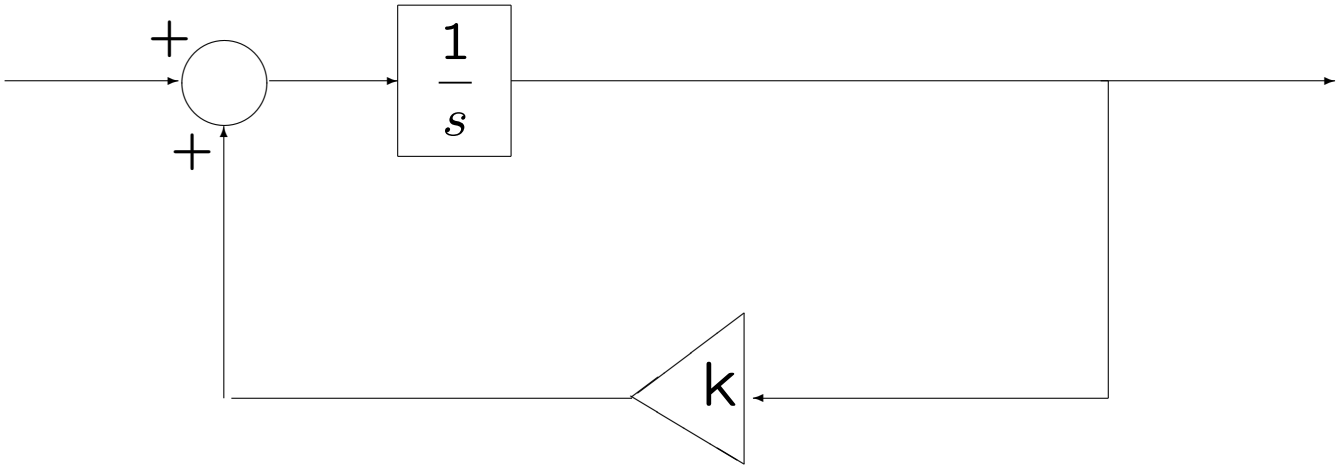


Takeoff Point



Scalor or Gain Multiplier




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Deriving States from Diff. Eqs. and Block Diagrams

- Continuous states can be identified as outputs of integrators
- Discrete states can be identified either as sampled continous states or as outputs of delayors

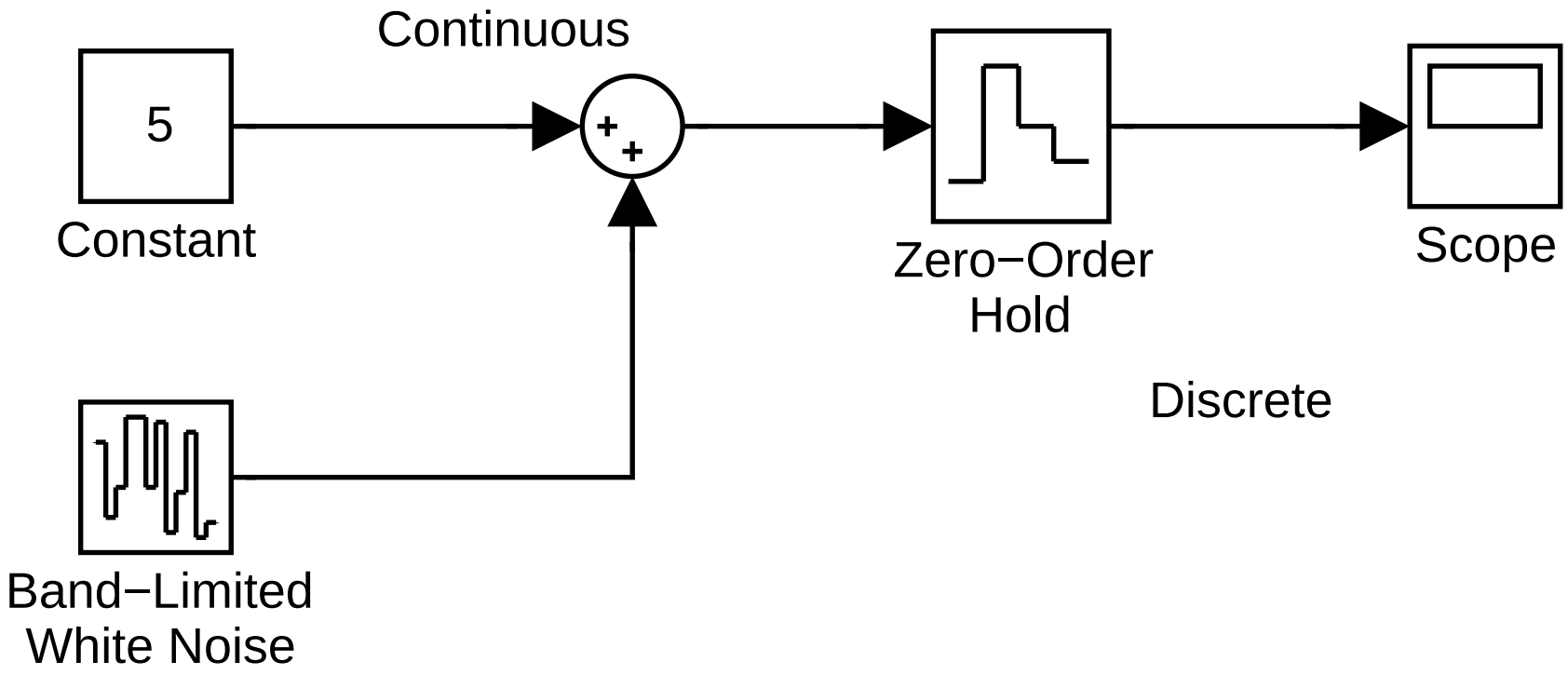
Examples

- Scalar Constant Dynamics
- Scalar Constant Dynamics with Recursive Estimator
- Scalar Ramp Dynamics with Recursive Estimator
- Damped Harmonic Oscillator

Scalar Constant Dynamics

- Matlab Model
- Very Simple
- Additive Noise
- No feedback loops

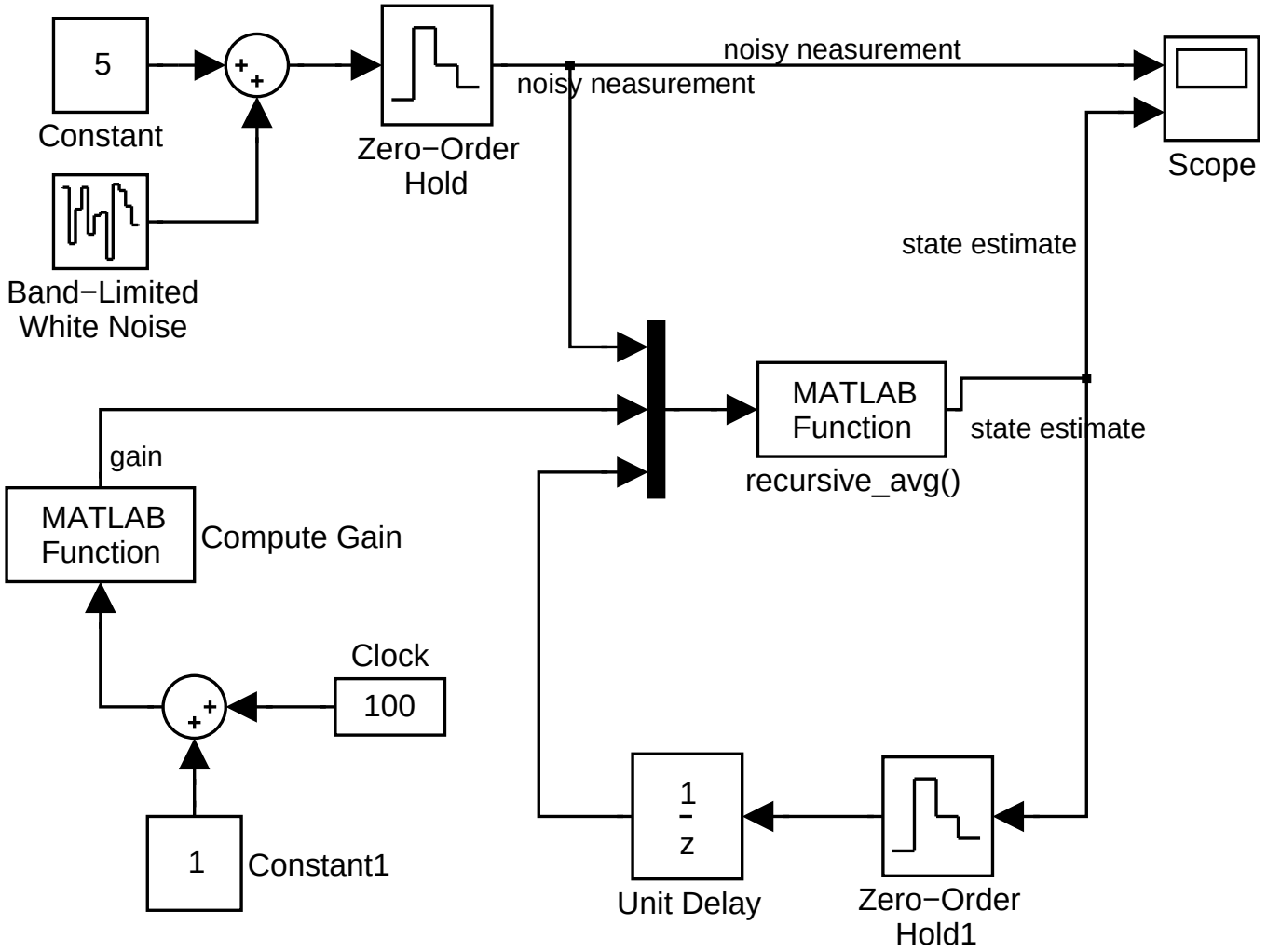
Scalar, Constant Dynamics plus White Process Noise



Scalar Constant Dynamics with Recursive Estimator

- Matlab Model
- Very Simple Dynamics
- Additive Noise
- Estimation loop implementing recursive simple averaging
- Simple open loop gain input

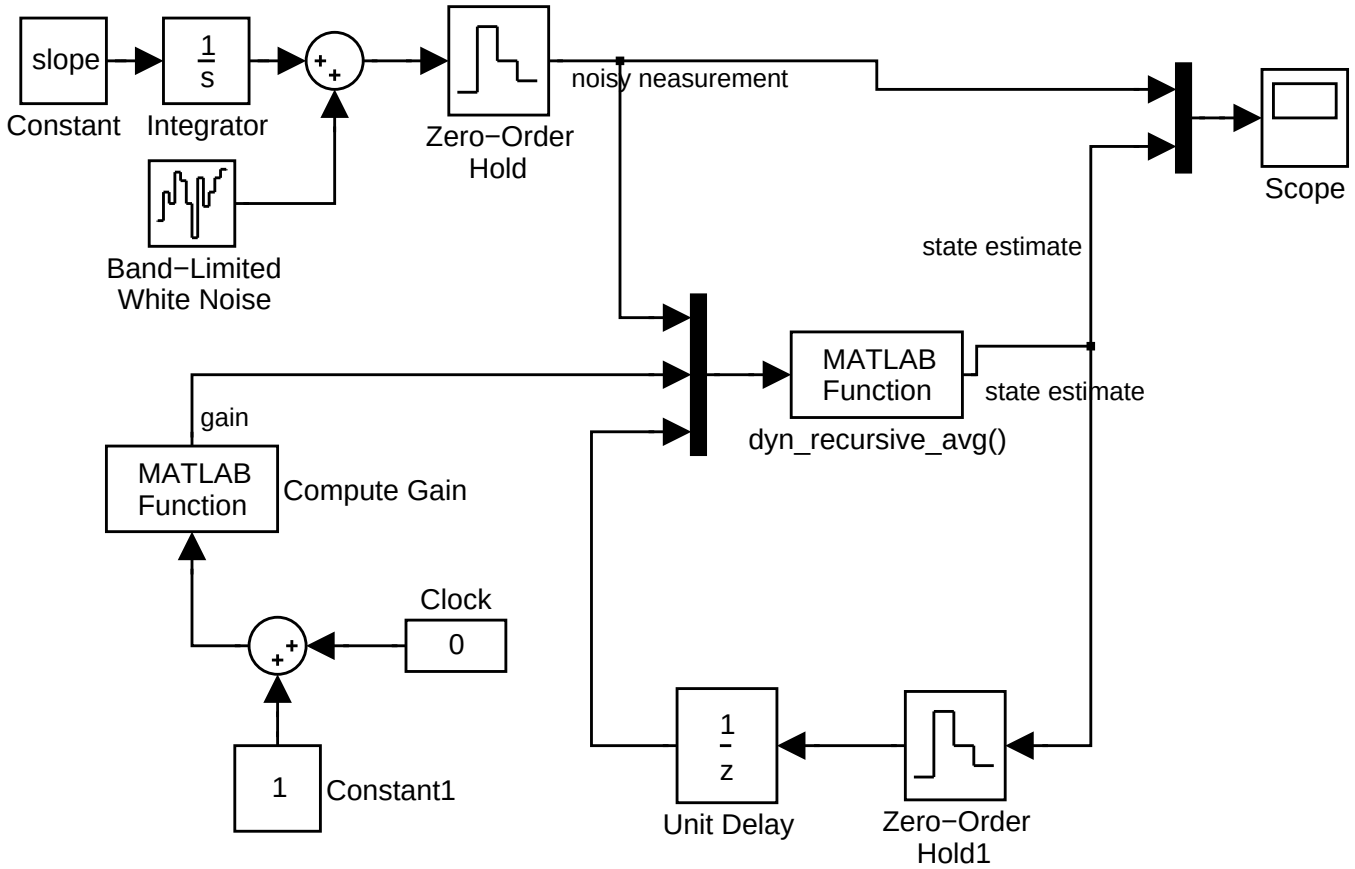
Scalar, Constant Dynamics plus White Process Noise with Estimation



Scalar Ramp Dynamics with Recursive Estimator

- Matlab Model
- Scalar Dynamics with known ramp
- Additive Noise
- Estimation loop implementing recursive simple averaging
- Simple open loop gain input

Scalar, Ramp Dynamics
plus White Process Noise
with Dynamic Average Estimation



Observe how the sampling rate affects the ability of the filter to follow the input.

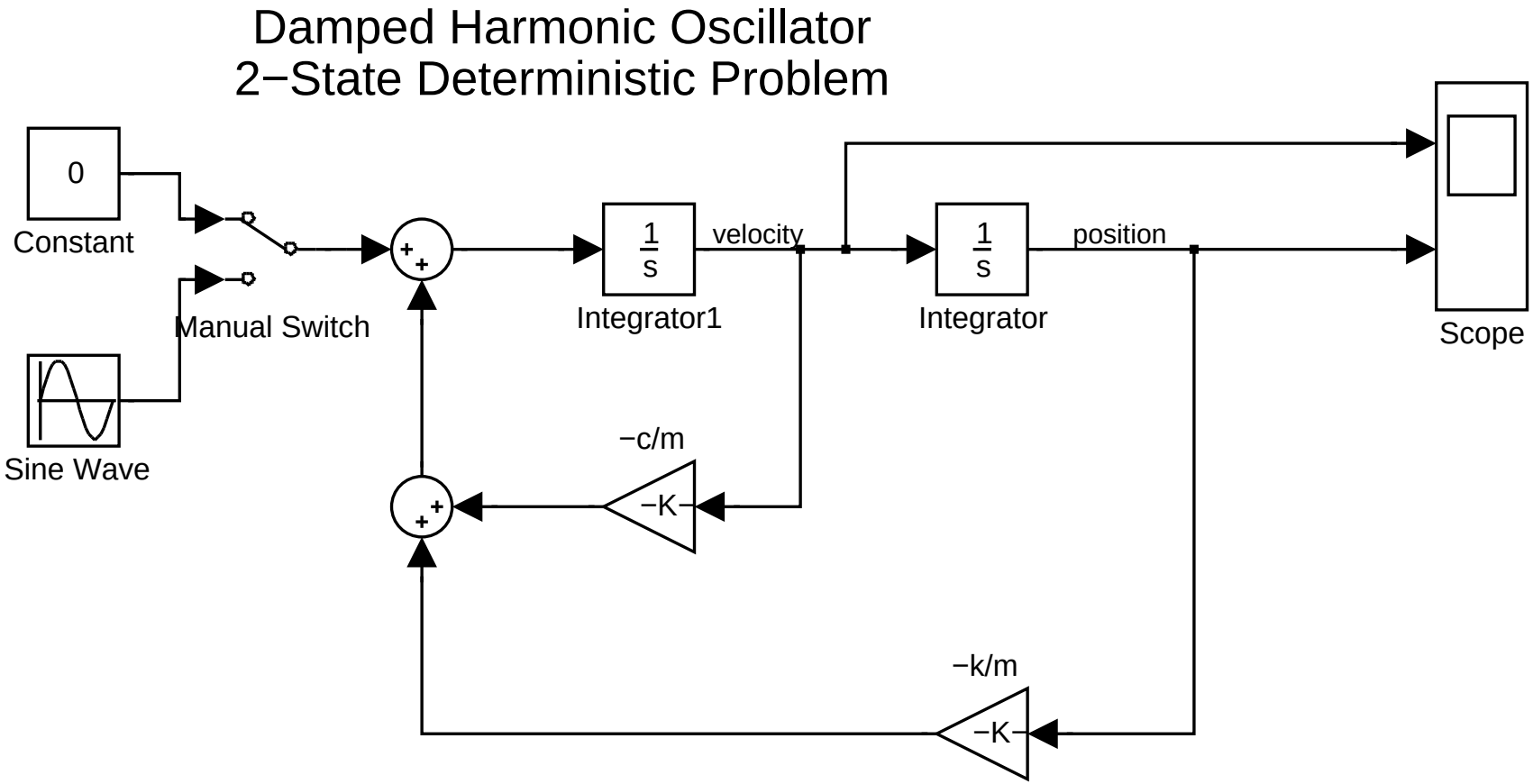
Vector Dynamics: Damped Harmonic Oscillator

- Matlab Model

- 2-dimensional Vector Dynamics

- Deterministic, Continuous Model

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -k/m & -c/m \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$


Damped Harmonic Oscillator
2−State Deterministic Problem
0
Constant
Manual Switch
Sine Wave
1
s
Integrator1
velocity
1
s
Integrator
position
Scope
−c/m
−K
−k/m
−K

Vector Dynamics: Damped Harmonic Oscillator

- Discrete model would look like this

$$\begin{bmatrix} x_1(k) \\ x_2(k) \end{bmatrix} = \exp\left(\begin{bmatrix} 0 & 1 \\ -k/m & -c/m \end{bmatrix} \Delta t \right) \begin{bmatrix} x_1(k-1) \\ x_2(k-1) \end{bmatrix}$$

- Here $\Delta t = t_k - t_{k-1}$

- Note that exp is a matrix exponential.

- In the continuous case, the state transition matrix is a matrix exponential whenever the system dynamics matrix is a constant (i.e., time invariant) matrix.

State Transition Matrix

- $\Phi_k = \Phi(k, k-1)$
- Solution of homogeneous equation for continuous systems
- If discrete system is derived from continuous system, then Φ_k satisfies:

$$x_k = \Phi_k x_{k-1}$$

Exercises

1. Convert the differential equation

$$5\frac{d^3y}{dt^3} - 0.3\frac{dy}{dt} + 7y = u(t)$$

   into a state space system model

2. Convert the difference equation

$$100y(k+9) = \sum_{i=0}^{2} b_i u(k+2-i)$$

3. Write the output equations for the above systems as well!